

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP9 - Enterprise Risk Management

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q.1)	
i)	Top down approach: • Company is new and Board's buy-in should be obtained • First time implementation may be more structured this way for the company Adv: This is more suitable with the company adopting ERM for the first time. Aligned Strategic objectives: • ERM should align with the company's mission to support SMEs. • ERM helps enhance decision-making and organizational resilience. Adv: The ERM should align with the vision of the company of supporting the SMEs and the strategy should be in alignment thereby. Designing the ERM program: • Keeping the program simple • Straightforward to manage Adv: simple ERM for this company may keep the Board and management motivated and interested. Risk Appetite and Tolerance • Define and communicate the company's risk appetite and tolerance levels. • Ensure that all business decisions align with these risk thresholds. Adv: The ERM should be based on the level of risk the company may bear and thresholds given the complexities of banking and financial services dynamics. Utilise the current risk management in the business units: • To adopt the best practices from the risk management in the respective business units and departments • Learn from the outcomes from the existing risk management Adv.: the risk management in in formal or informal form in current lines of business or units may be refined rather than reinventing the wheel for such successfully working methods. Risk culture and communication about ERM: • Clearly communicate the benefits and gains from the ERM implementation • Outline briefly the role of stake holders involved Adv.: the communication of benefits of ERM program increases the acceptability of the same. Risk Governance and Accountability • Clear governance structure with defined roles for the Board, senior management, and business units. • Ensure accountability for identifying, assessing, and mitigating risks at all organizational levels. Adv.: It helps in defining the roles and responsibilities of all involved in the ERM process. Compliance with Regulatory Requirements • Ensure the ERM program includes controls to comply with evolving AML and other regulatory norms. • Mitigate risks related to non-compliance, reducing the chance of penalties or reputational harm. Adv.: the new regulatory provisions on anti - money laundering are important to comply and underlying risks to be reduced accordingly. It may otherwise pose regulatory risks and attract regulatory fines. Collaboration and Stakeholder Engagement

	- Engage with internal and external stakeholders, including regulators and industry experts. - Promote cross-functional collaboration to strengthen risk management practices across the organization. Adv.: The involvement of stakeholders and keeping them in confidence is important as the areas of operation of the company and its customer base may be sensitive to the ERM. Comprehensive Risk Identification - Holistically assess financial, operational, technological, regulatory, and reputational risks. - Focus on key areas like fraud prevention, cyber risks, and regulatory compliance. Adv.: Given the high risk of recent frauds and cyber incidents, the detailed risk identification becomes very crucial. (15)
ii)	- Defining risks and ERM - Advantages of ERM over Risk management - Examples of impact of downside risks in companies in similar domain e.g. cyber, credit and operational risks - Steps of risk management framework - Roles, responsibilities and governance, monitoring and feedback systems - Risk-heat map and other approaches of risk management (Max 3)
iii)	1. Strategic Risks: - Strategy Implementation: Challenges in effectively implementing the company's strategic objectives, including expanding operations or introducing new services. - Depositor Demographics: The company's predominantly high-age customer base poses a concentration risk and limits future growth potential among younger demographics. - Competition: Increasing competition from other financial institutions or fintech companies could threaten market share and profitability. 2. Credit Risks: - Default Risk: The risk of borrowers, particularly small and medium enterprises (SMEs), failing to meet their repayment obligations. - Concentration of Lenders: A high concentration of loans to a specific segment or industry could expose the company to higher default risks during economic downturns. 3. Financial Risks: - Liquidity Risk: The risk of not having enough liquid assets to meet obligations such as customer withdrawals or payments. - Capital Management: Ensuring the company maintains sufficient capital to meet regulatory requirements and sustain operations during economic stress. 4. Operational Risks: - Information Technology: Risks related to outdated or insufficient IT infrastructure that could impact service delivery or data management. - Information Security: The risk of data breaches, loss of sensitive information, and inadequate protection against cyber threats. - Fraud: The risk of internal or external fraud, including financial misreporting, embezzlement, or fraudulent transactions.

- Outsourcing: Risks associated with outsourcing operations, including vendor management and the quality of outsourced services.
- Personnel: Risks related to workforce management, employee errors, or lack of skilled personnel in critical areas.
- Cyber Threats: Exposure to cyber-attacks, phishing, ransomware, and other technological risks that could lead to financial or reputational damage.

5. Compliance Risks:

- Regulatory Compliance: The risk of non-compliance with local and international regulations, particularly related to anti-money laundering (AML) laws and customer due diligence.
- Money Laundering: The potential for the company to be used as a channel for illicit financial transactions, leading to regulatory penalties and reputational damage

(Max 7)

iv)

<i>Risk Category</i>	Operational	Operational/Regulatory	Financial	Strategic
<i>Sub- Risk Category</i>	IT risk	Compliance	Credit Risk	Environment concentration
<i>Risk Description</i>	Cyber incident	Money Laundering	Industry concentration and counter party risk	Demographic risk due to ageing customers
<i>Impact scale</i>	Extreme	Major	Moderate	Major
<i>Likelihood scale</i>	Possible	Likely	Probable	Likely
<i>Inherent Risk scale</i>	Critical	Major	Major	Major
<i>Control mechanism</i>	- Latest system patches and security solutions - Cyber assurance audit and gap assessment	Anti-Money laundering awareness and trainings	- More efficient underwriting on property loans - Predictive and business insights on future trends	- Balancing of age-distributed customers' mix in the portfolio - More effective advertising campaigns
<i>Residual risk</i>	Major	Moderate	Moderate	Major
<i>Responsible team</i>	- Information security team and officer - Chief Risk Officer	Chief Compliance and Legal team	Chief Financial Officer	Marketing and Sales team

Likelihood scales:

Rating	Likelihood	Frequency
Probable	- 70% in a year	Will likely happen once in a year
Likely	- 40-70% in a year	May likely happen once in next 5 years
Possible	- 10-40% in a year	May likely happen once in next 5 -10 years

	<table><tr><td>Unlikely</td><td>< 10% in a year</td><td>Rare, may happen in next 10 years or more</td></tr></table>	Unlikely	< 10% in a year	Rare, may happen in next 10 years or more									
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	Severity scales: <table><tr><td>Rating</td><td>Financial Impact</td><td>Regulatory Impact</td></tr><tr><td>Major</td><td>- Dip in revenue by 10-20%</td><td>Regulatory fines/penalties</td></tr><tr><td>Moderate</td><td>- Dip in revenue by 5-10%</td><td>Regulatory formal warning</td></tr><tr><td>Minor</td><td>- Dip in revenue by < 5%</td><td>Regulatory</td></tr></table> (Max 10) [35 Marks]	Rating	Financial Impact	Regulatory Impact	Major	- Dip in revenue by 10-20%	Regulatory fines/penalties	Moderate	- Dip in revenue by 5-10%	Regulatory formal warning	Minor	- Dip in revenue by < 5%	Regulatory
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Q. 2)													
i)	- User enters the pin code, name of region or any other relevant identifier of the manufacturing unit under study, into the model. - The model codifies the provided location into the latitude and longitude or the transformed locational co-ordinates. - The stochastic function defines the event occurrence for the location and cyclone. - Hazard module generates information including the wind speed and approach of cyclonic storm to determine the severity of the cyclonic effects - The vulnerability function determines the mean impact or damage and the residual impact on the manufacturing unit. The impact also considers the characteristics of the factory e.g. area, vulnerable zones, safety hazards etc - Financial function estimates the losses using the estimated average damage, based on the factory's value and the extent of coverage through insurance. - With the deviation in estimation and the distribution parameters taken into account, financial loss is quantified for the lines of business and the respective coverages that the manufacturing unit might have had. (Max 6)												
ii)	1. Importance of Data Quality in Cyclone Models • Poor data quality impacts cyclone model accuracy and leads to discrepancies between modelled and actual losses. • Poor data can account for significant gaps in loss estimates in cat models' Actual vs Expected. 2. Factors Affecting Modelled Losses • High-resolution, location-specific data (latitude, longitude) is key for accurate results. • Building characteristics (construction type, occupancy, year built) reduce uncertainty in loss distributions. 3. Consequences of Poor-Quality Data • Incorrect coding of risk types skews loss estimates. • Using coverage limits or missing building data can distort modelled results.												

	• Incomplete or inaccurate data increases loss distribution spread. 4. Boardroom Relevance and Industry Implications • Data quality impacts financial strength ratings and regulatory capital requirements. • Rating agencies scrutinize data quality, influencing premiums and capacity. 5. Value of Enhanced Data Quality • Better data reduces operational risk and improves decision-making. • Higher data quality improves financial ratings and access to appropriate capital. • The company being a Re-insurer may offer premium credits for companies with strong data controls. 6. Challenges in Improving Data Quality • Difficulties in setting objectives and benchmarking improvements. • Companies must weigh costs and benefits of data improvement initiatives. 7. Road Ahead for Cyclone Models • Companies may benchmark current data quality for improvement, which may be explored to use to add value to the cat models. • Strong data controls offer competitive advantages. (Max 8)
iii)	- ROE and ROA focus relative to balance sheet items. - ROA and ROE focus on the assets, while only ROA and not ROE considering the impact of leverage. - Both ROE and ROA do not consider the riskiness of the asset and the return resulting from accounting for the risk taken. - RAROC and RARORAC are risk-adjusted measures. - RAROC and RARORAC are measures of profitability of the portfolio. - RAROC and RARORAC may be of more utility for the comparison between portfolio and insurers. - RAROC may be of interest to the shareholders as it can convey about the return on capital infused by them. - RARORAC may be of utility to the internal management as it also considers risk adjusted returns on the capital required to generate it. - RAROC and RARORAC may be measured at business or lines of businesses. - The result of these two measures may give an indication towards performance of the product/line of business w.r.t. the capital requirement in the same. - These may also give some inference on which product/line of business may be more capital intensive to generate a certain risk-adjusted return and those which may have some scope of freeing up of capital. (Max 7)
iv)	$EC = A - A \{ (1 + r_a) * (1 - EL) / (1 + r_i) \}$ Without diversification, fall in the portfolio = $0.65 * 21\% + 0.35 * 12\% = 17.85\%$ $EC-1 = 150 - 150 * (1.0723) * 0.8215 / 1.0532 = 24.54$ With diversification:

	EC-2 = 150 – 150 *(1.0723) *(1-14.4%)/1.0532 = 19.28 %age difference in EC with and without diversification is: 21.47% i.e. the EC with diversification is 21.47% lower than that with diversification. (3)																
v)	EC-A = 0.65 * 150 – 0.65 * 150 *(1.0729) *0.79/1.0532 = 19.03 EC-B = 0.35 * 150 – 0.35 * 150 *(1.0711) *0.88/1.0532 = 5.51 Proportion of allocation of asset A = 77.5% Proportion of allocation of asset A = 22.5% Allocation of Capital to Asset A: 77.5% of EC-2 (as in the previous sub-part) = 14.95 Allocation of Capital to Asset B: 22.5% of EC-2 (as in the previous sub-part) = 4.33 Summary of Balance sheet at the start of year: <table><tr><td></td><td>A</td><td>B</td><td>Total</td></tr><tr><td>Asset</td><td>97.5</td><td>52.5</td><td>150</td></tr><tr><td>Liability</td><td>82.55</td><td>48.17</td><td>130.72</td></tr><tr><td>EC</td><td>14.95</td><td>4.33</td><td>19.28</td></tr></table> (Max 8)		A	B	Total	Asset	97.5	52.5	150	Liability	82.55	48.17	130.72	EC	14.95	4.33	19.28
	A	B	Total														
Asset	97.5	52.5	150														
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vi)	The assets grow at 7.29% and 7.11% respectively and there is impact of the expected losses at 1.45% and 1.26% respectively. Therefore, at the end of year: Value of asset-A = 97.5 * (1.0729)*(1-0.0145) = 103.09 Value of asset-B = 52.5 * (1.0711)*(1-0.0126) = 55.52 Liability – A = 82.55 *(1.0532) = 86.94 Liability – A = 48.17 *(1.0532) = 50.73 EC-A = 16.15 EC-B = 4.79 RARORAC – A = 16.15/14.95 – 1 = 8.02% RARORAC – A = 4.79/4.33 – 1 = 10.66% It is observed that in spite of the asset A having higher expected return may be a less attractive instrument on risk-adjusted basis. This is because asset based on the above calculation appears to be requiring more capital given due to its higher risk in view of the risks and expected losses. (Max 8) [40 Marks]																
Q. 3)																	
i)	Approach for quantification for cyber Risk assessment:																

	- understanding the weaknesses in IoT devices - Classification of risks in to appropriate lines of business - Analysis of dependencies and loss distributions - Pricing using projection of estimated losses - Support of reinsurance or other sources of data to be explored - Review of performance of such products in other markets and global experience Risk considerations: - Tail dependencies to be considered in such losses as it may impact multiple lines of business - Single losses can large and may require high capital allocation Actuarial approaches: - Large losses may be modelled separately for the input of estimated losses into pricing - Pricing with deductibles may be considered which may help in adverse selection - Suitable capital allocation margins may be set considering the dependency structure Challenges: - There may be lack of sufficient volume and credibility of data to be utilised for modelling for pricing - Individual assessments for each case in hand for underwriting may be considered to price in initial stages, which may require higher underwriting efforts and costs - Regulatory provisions on such products, ransom payments may impact design of such products (Max 7)
ii)	- Model validations have produced important and major findings in the past - There are actuarial team members who are exclusively focussed and dedicated towards the model validation. - They are independent of model development and ensures the independence and no bias in their review functions. - The model review at the multiple layers adds value in terms of maintenance aspects, conceptual clarity and design of models. - Such reports also reach the senior management who can regularly review the same and this helps them in decision making. - Informed decisions on documentations, strengthening various aspects of model development, allocation of resources can be assisted with such approaches. - The cost and impact of any potential errors which may otherwise result are significantly minimised. - Model changes may be thoroughly reviewed with multi-line of validations which reduces the sensitivity of the changes. - Such deep layers of reviews can reduce the load of work with internal or external auditors. - Roles for each line of defence may be specified to avoid duplication of efforts, such as co-ordination with other oversight function such as internal audit, other to be risk-based. (Max 7)
iii)	- Modelling team should model and document all the changes whether minor or major, meticulously. - Test the changes on stand-alone basis as well as aggregate level.

	- Quantify the impact of each of the changes. - Consider the impact of correlation in estimating the impact of such changes - Reasonableness check and balances to be constantly assessed - Categorise the changes in case the number of changes is too high and attribute the impact to each category - Offsetting impact may be approached from a top-down way by starting with the study of impact on a certain category - The changes are to be reviewed independently by the subsequent layers of defence in the model review process - Changes to be presented to the any relevant committee of model governance - Additional peer review may also be carried out in case the estimated impact is higher than the threshold prescribed by the company's policy - Rules and controls around the model validation and review process to be regularly reviewed - Potential future errors and learning from the past changes' impact any offsetting to be kept in consideration in any process (Max 6)
iv)	In the (re)-designed model validation framework: - The 1st line/layer of defence needs to have adequate controls - Preventive measures and internal checks and balances are to be inbuilt into the models - Current documents to be enhances with such findings - Model inputs to be undergo independent tests for reasonableness w.r.t. the product designs. The 1st line of defence should have continuous monitoring of the model's outputs - This may include time-horizon based analysis of outputs vis-à-vis model inputs - To ensure that the differences are explainable - Spurious accuracy to be checked at all stages of development - Inconsistency in outputs or inputs to be analysed w.r.t. each assumption also - Various ratios w.r.t. products or lines of business may be considered - Boundary conditions and outliers to be established wherever possible in the model development cycle. (Max 5) [25 Marks]
